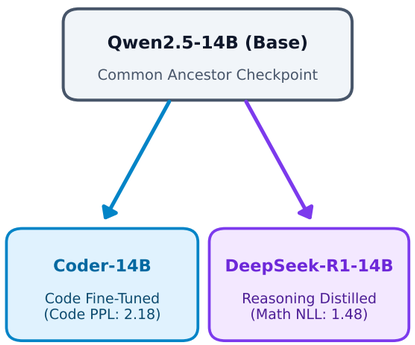
Phylogenetic Tree Surgery on 14B LLMs (Qwen2.5-14B, Coder-14B, DeepSeek-R1-14B)

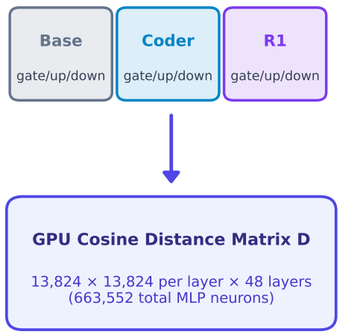
From 663,552 MLP Neurons → 48 GPU BME Layer Trees (6,171 Clades) → Quartet Topology Violations → Surgical Knock-Out & Zero-Training Chimera

STEP 1 • GENOMES — Sibling Models & Signatures

Three Sibling 14B Checkpoints (Same 48-Layer × 13,824-Neuron Body)



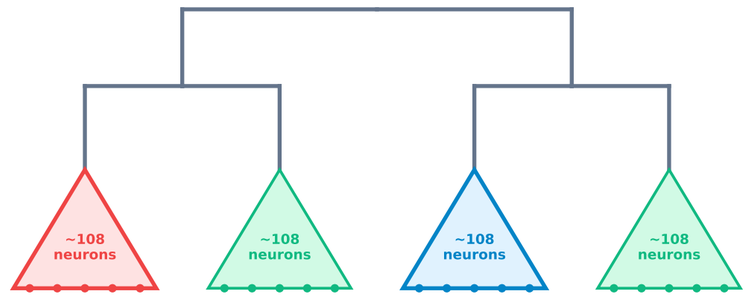
Each MLP Neuron i = One 'Species' (13,824 Neurons per Layer)



Homology across Checkpoints: Because all 3 models share the same Base initialization, neuron i in layer l is homologous across Base, Coder, and R1 — enabling phylogenetic trees!

STEP 2 • TREES — GPU BME Trees (6,171 Clades)

GPU Balanced Minimum Evolution (BME) Tree per Layer
13,824 Neurons → ~128 Clades/Layer (6,171 Total Clades)



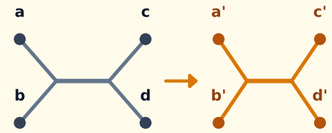
Clade #14 (Violated) Rewired for Math ΔQ = 0.41 (High)
Clade #42 (Conserved) Shared Syntax ΔQ = 0.02 (Low)
Clade #89 (Violated) Rewired for Code ΔQ = 0.38 (High)
Clade #112 (Conserved) Core Knowledge ΔQ = 0.01 (Low)

Phylo2Vec GPU Discrete Gradient SPR infers each 13,824-neuron tree in 1.8s (<90s total)

STEP 3 • QUARTETS — Partner Swaps vs. Weight Norms

1. Uniform Subspace Drift

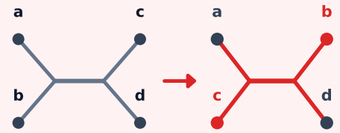
(Fools ||ΔW||_F & TIES-Merging)



- Weights rotated (large ||ΔW||_F)
- Split {a,b}|{c,d} is UNCHANGED
- Quartet Violation = 0 (Ignored!)

2. True Circuit Rewiring

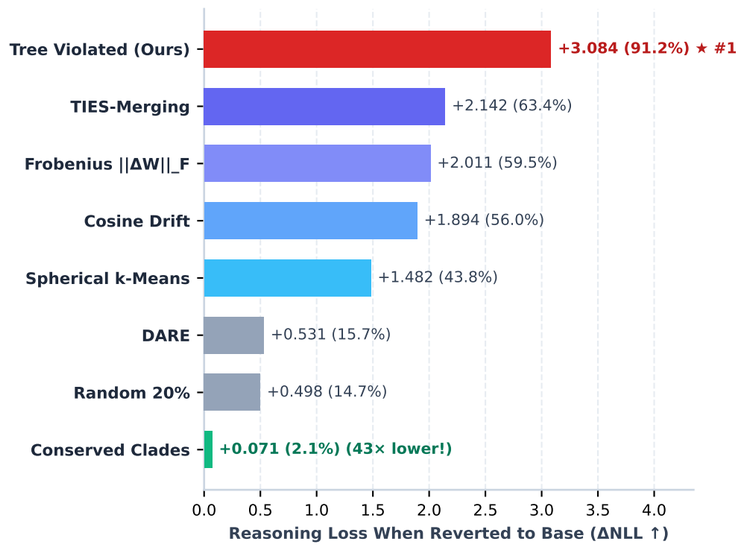
(Caught by Quartet Violation)



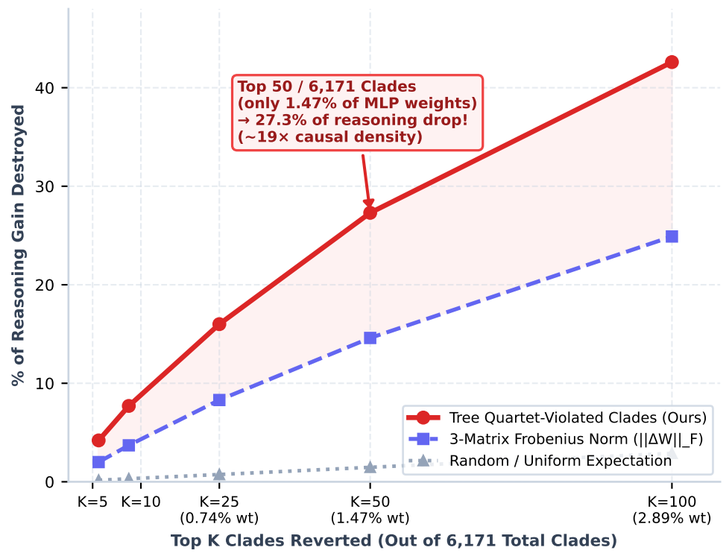
- **Neurons b & c SWAPPED partners: {a,b}|{c,d} → {a,c}|{b,d}!**
- **Relational geometry broke (High ΔQ)**

1-Body Norms ask: 'Did neuron i move?' vs. Tree Surgery asks: 'Did this clade change its partners?'

RESULT 1 • ABLATION — Beats 7 Baselines (20% Budget)



RESULT 2 • SPARSITY — Top 50 Clades = 27.3% of Effect



RESULT 3 • CHIMERA — Zero-Training Specialist Grafting

